Notations :

1.Options shown in green color and with ✔ icon are correct.

2.Options shown in red color and with ✖ icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

Speech Technology

Section Id :	640653139209
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	25
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653389911
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406532135837 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SPEECH TECHNOLOGY (PEN AND

PAPER EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536956542. ✔ Yes

6406536956543. ✖ No

Question Number : 2 Question Id : 6406532135838 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

Instructions :

All **option-based questions are MCQs,** and all **fill-in-the-blank questions are short answer questions.**

For Questions **3 to 20,** answer all questions.

• In **MCQs,** you must write the **correct option** and also the **complete text of that option.**
• You must then provide a **detailed explanation** of why that specific option is the correct answer.
• In **short answer questions,** write the answer clearly with proper explanation wherever needed.
• In all **problem-solving or example-based questions,** present the solution with **all intermediate steps clearly shown.**
• Step-wise explanation is **compulsory** for all such questions.
• The **final answer** must be written clearly at the end.

Options :

6406536956544. ✔ Instructions has been mentioned above.

6406536956545. ✖ This Instructions is just for a reference & not for an evaluation.

Sub-Section Number :	2
Sub-Section Id :	640653389912
Question Shuffling Allowed :	No

Question Number : 3 Question Id : 6406532135839 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If a speech signal is sampled at 16,000 samples per second, how many samples are contained in a 10-millisecond frame?

Options :

6406536956546. ✔ I have answered this question in the offline paper

6406536956547. ✖ I have skipped this question

6406536956548. ✖ This question does not apply to me

Question Number : 4 Question Id : 6406532135840 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How does HuBERT differ from Wav2Vec 2.0 in target generation?

(a) Uses labeled phonemes
(b) Uses online quantization only.
(c) Uses offline clustering (e.g., k-means).
(d) Does not use masking.

Options :

6406536956549. ✔ I have answered this question in the offline paper

6406536956550. ✖ I have skipped this question

6406536956551. ✖ This question does not apply to me

Question Number : 5 Question Id : 6406532135841 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Why is ”deduplication” a common step when tokenizing speech signals into phonetic tokens?

(a) To remove noise from the original audio recording.
(b) To account for the fact that speech units (like vowels) exist for multiple time frames (e.g., 70-100ms) and map to the same cluster ID.
(c) To increase the vocabulary size of the LLM.
(d) To convert continuous SSL embeddings into discrete vectors.

Options :

6406536956552. ✔ I have answered this question in the offline paper

6406536956553. ✖ I have skipped this question

6406536956554. ✖ This question does not apply to me

Question Number : 6 Question Id : 6406532135842 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

In a neural codec framework using Residual Vector Quantization (RVQ),how many codewords are typically generated per vector?

(a) Number of RVQ layers
(b) Size of each codebook
(c) Number of encoder layers
(d) Sampling rate of the audio

Options :

6406536956555. ✔ I have answered this question in the offline paper

6406536956556. ✖ I have skipped this question

6406536956557. ✖ This question does not apply to me

Question Number : 7 Question Id : 6406532135843 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

You are training a Speech LLM. Your audio codec outputs 4 RVQ codebooks.The frame rate is 50 Hz. If you choose a strictly flattened Autoregressive (AR) decoding strategy (predicting layer 1, then layer 2, then layer 3, then layer 4 for frame 1, before moving to frame 2), what is the total sequence length of tokens the LLM must process for a 5-second audio clip?

(a) 250 tokens
(b) 1000 tokens
(c) 1250 tokens
(d) 5000 tokens

Options :

6406536956558. ✔ I have answered this question in the offline paper

6406536956559. ✖ I have skipped this question

6406536956560. ✖ This question does not apply to me

Question Number : 8 Question Id : 6406532135844 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Imagine that you design a closed-loop system where speech is synthesized from text (TTS) and then transcribed back into text (ASR). Which evaluation measure best captures the accuracy of the text reconstruction at the end of this loop?

(a) Signal-to-noise ratio (SNR) of the reconstructed audio
(b) Cosine similarity between mel-spectrograms of original and reconstructed audio
(c) Perplexity of the reconstructed text under a language model
(d) Word Error Rate (WER) between the original and reconstructed text

Options :

6406536956561. ✔ I have answered this question in the offline paper

6406536956562. ✖ I have skipped this question

6406536956563. ✖ This question does not apply to me

Question Number : 9 Question Id : 6406532135845 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

In the CELP algorithm, what does the ”Codebook index” specifically represent?

(a) The shape of the vocal tract
(b) The index of a pre-defined excitation signal that best matches the residual
(c) The volume level of the speaker

(d) The duration of the phoneme

Options :

6406536956564. ✔ I have answered this question in the offline paper

6406536956565. ✖ I have skipped this question

6406536956566. ✖ This question does not apply to me

Question Number : 10 Question Id : 6406532135846 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What is the mathematical advantage of RVQ over standard VQ for a high-dimensional vector space?

(a) RVQ eliminates the search time for the closest centroid
(b) VQ requires K^N codebook entries for the same precision that RVQ achieves with $K \times N$ entries
(c) RVQ ensures the reconstructed vector is always mathematically identical to the original
(d) RVQ works better with non-linear activation functions in the encoder

Options :

6406536956567. ✔ I have answered this question in the offline paper

6406536956568. ✖ I have skipped this question

6406536956569. ✖ This question does not apply to me

Question Number : 11 Question Id : 6406532135847 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Which system is most likely to fail if the same sentence is spoken by two different speakers with identical accents?

(a) ASR
(b) Speaker Verification
(c) TTS
(d) MT

Options :

6406536956570. ✔ I have answered this question in the offline paper

6406536956571. ✖ I have skipped this question

6406536956572. ✖ This question does not apply to me

Question Number : 12 Question Id : 6406532135848 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Why is predicting the immediate next frame (e.g., at t+1) often considered a ”trivial” task in speech processing?

(a) Speech signals are completely random.
(b) Digital speech has very low sample rates.
(c) Closely occurring frames (10ms apart) in speech are highly correlated.
(d) Next-token prediction is only effective for text, not audio.

Options :

6406536956573. ✔ I have answered this question in the offline paper

6406536956574. ✖ I have skipped this question

6406536956575. ✖ This question does not apply to me

Question Number : 13 Question Id : 6406532135849 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

In RNN-T, if the Joiner network uses a simple addition operation $z_{t,u} = \phi(h_t^{enc} + h_u^{pred})$, what happens if the Encoder output h_t^{enc} becomes zero?

(a) The model becomes a pure CTC model.
(b) The model behaves like a pure unconditional Language Model.
(c) The model fails to converge.
(d) The output becomes random.

Options :

6406536956576. ✔ I have answered this question in the offline paper

6406536956577. ✖ I have skipped this question

6406536956578. ✖ This question does not apply to me

Question Number : 14 Question Id : 6406532135850 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If we convert speech to pure text, what information is primarily lost?

(a) The literal words spoken
(b) The grammatical structure
(c) Speaker identity and emotion
(d) The language of the speaker

Options :

6406536956579. ✔ I have answered this question in the offline paper

6406536956580. ✖ I have skipped this question

6406536956581. ✖ This question does not apply to me

Question Number : 15 Question Id : 6406532135851 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The SUPERB challenge is a benchmark specifically designed to evaluate:

(a) Only Automatic Speech Recognition (ASR) models
(b) Self-supervised models across multiple speech tasks
(c) The physical hardware used for training
(d) Text-only LLM performance

Options :

6406536956582. ✔ I have answered this question in the offline paper

6406536956583. ✖ I have skipped this question

6406536956584. ✖ This question does not apply to me

Sub-Section Number : 3

Sub-Section Id : 640653389913

Question Shuffling Allowed : No

Question Number : 16 Question Id : 6406532135852 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Contrast the architectural paradigms of Wav2Vec 1.0 and Wav2Vec 2.0. How does the shift from a strictly autoregressive approach (Wav2Vec 1.0) to a bidirectional masked approach (Wav2Vec 2.0) impact the contextual representations learned?

Options :

6406536956585. ✔ I have answered this question in the offline paper

6406536956586. ✖ I have skipped this question

6406536956587. ✖ This question does not apply to me

Question Number : 17 Question Id : 6406532135853 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

When applying K-Means clustering for speech discretization, a specific vocabulary size (number of clusters, K) must be defined. Discuss the trade-offs between choosing a very small vocabulary (e.g., K=50) versus a very large vocabulary (e.g., K=2000). How does this choice impact phonetic discriminability and token sequence bit-rate?

Options :

6406536956588. ✔ I have answered this question in the offline paper

6406536956589. ✖ I have skipped this question

6406536956590. ✖ This question does not apply to me

Question Number : 18 Question Id : 6406532135854 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Given a latent vector: z=[1.2,0.8] and codebook vectors: c1=[1,1], c2=[0,0]

(a) Find the closest codebook vector.
(b) Compute the quantized output.

Options :

6406536956591. ✔ I have answered this question in the offline paper

6406536956592. ✖ I have skipped this question

6406536956593. ✖ This question does not apply to me

Sub-Section Number : 4

Sub-Section Id : 640653389914

Question Shuffling Allowed : No

Question Number : 19 Question Id : 6406532135855 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Describe the process of modality alignment in a SpeechLLM incase of:

(a) SpeechGPT
(b) SLAM-ASR

Explain with a simple architecture diagram.

Options :

6406536956594. ✔ I have answered this question in the offline paper

6406536956595. ✖ I have skipped this question

6406536956596. ✖ This question does not apply to me

Question Number : 20 Question Id : 6406532135856 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Describe the process of how a multi-layer 1D CNN block reduces the sequence length of high-frequency raw audio (e.g., 16 kHz) into a manageable sequence of latent representations. A single 1D CNN layer in a feature extractor has a kernel size of K=10, a stride of S=5, and zero padding P=0. If the input raw audio segment consists of I=400 samples, calculate the temporal dimension, O of the output feature map using the standard convolution formula.

Options :

6406536956597. ✔ I have answered this question in the offline paper

6406536956598. ✖ I have skipped this question

6406536956599. ✖ This question does not apply to me